Progress Report

CMM-VAE for Winter Precipitation Regimes over Morocco

MSWEP Dataset · DJF Seasons · 1979–2020

April 2026

Dataset	MSWEP precipitation + ERA5 Z500
Period	DJF winters 1979/80 – 2019/20 (42 winters)
Model	CMM-VAE (Coupled Mixture Model VAE)
Baseline	K-Means clustering on Z500
Notebooks	`cmmvae_mswep_djf_winter_groups.ipynb` `baseline_kmeans_djf.ipynb`

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Overview

This report summarises the work done to evaluate the **Coupled Mixture Model Variational Autoencoder (CMM-VAE)** for identifying atmospheric circulation regimes associated with extreme winter precipitation over Morocco.

Two notebooks were developed and compared:

1. `cmmvae_mswep_djf_winter_groups.ipynb` – the main CMM-VAE model
2. `baseline_kmeans_djf.ipynb` – a K-Means clustering baseline

Both notebooks use **identical** data splits, preprocessing, and evaluation protocols to ensure a fair comparison.

Data

Table 1: Dataset summary

Item	Details
Atmospheric input (X)	ERA5 geopotential height at 500 hPa (Z500), Atlantic domain, 25×33 grid = 825 features
Precipitation (Y)	MSWEP daily precipitation over Morocco, spatially masked
Period	DJF winters 1979/80 – 2019/20 (42 winters)
DJF convention	December of year Y + January/February of year $Y+1$ = Winter Y

Key Methodological Change: Leakage-Free Winter-Group Split

The problem with naïve day-level splitting

A standard random train/test split applied to individual days can place **December 1990** in training and **January 1991** in the test set – days from the same meteorological winter. This causes **temporal data leakage**: the model is evaluated on data that is climatologically and dynamically correlated with its training data, making the evaluation over-optimistic.

The fix: GroupShuffleSplit on winter-year keys

A **winter-year group key** is assigned to every day:

$$\text{winter_year}(t) = \begin{cases} Y & \text{if month}(t) = \text{December} \\ Y-1 & \text{if month}(t) \in \{\text{January, February}\} \end{cases}$$

Example: Dec 1990, Jan 1991, Feb 1991 all receive winter_year = 1990.

`GroupShuffleSplit` (70 % / 30 %) then assigns **entire winters** to either the training or holdout set – a DJF season is never split across both sets.

Split result

Table 2: Train/holdout split summary

Set	Winters	Days (approx.)
Training	29 winters (70 %)	≈ 819
Holdout	13 winters (30 %)	≈ 353
Winter overlap	**0**	–

Training winters: 1978, 1979, 1980, 1983, 1985, 1987, 1988, 1989, 1991, 1992, 1993, 1994, 1995, 1996, 1997, 1998, 1999, 2000, 2001, 2004, 2006, 2007, 2008, 2010, 2011, 2013, 2016, 2018, 2019, 2020.

Holdout winters: 1981, 1982, 1984, 1986, 1990, 2002, 2003, 2005, 2009, 2012, 2014, 2015, 2017.

Preprocessing strictly from training data

To prevent any indirect leakage, all preprocessing is derived **exclusively from training days**:

- **KMeans precipitation labels** – fit on training days, then applied to holdout via `kmeans.predict()`
- **Q95 exceedance threshold** – 95th percentile computed from training pixels only, then applied unchanged to holdout days

Cross-validation respects winter boundaries

Inside the training set, **5-fold GroupKFold** ensures that no DJF winter is ever split between a CV training fold and a CV validation fold.

Model: CMM-VAE

The CMM-VAE is a Variational Autoencoder with a coupled mixture model that simultaneously:

1. **Encodes** Z500 circulation patterns into a latent space
2. **Clusters** them into K atmospheric regimes (soft assignment c_{enc})
3. **Predicts** the associated precipitation cluster (r_{enc})

Table 3: CMM-VAE architecture and hyperparameters

Parameter	**Value**
Input features	825 (Z500 anomaly, 25 × 33 Atlantic grid)
Encoder layers	825 → 256 → 128 → 64 → latent dim 15
Latent dimension	15
Decoder layers	15 → 64 → 128 → 256 → 825
Number of regimes (k)	4
Precipitation clusters	4 (KMeans on MSWEP, training only)
Activation	ReLU
Batch size	128
Epochs	100 (best fold selected)
Loss	Reconstruction (MSE × 0.5) + KL divergence + prediction loss
Cross-validation	5-fold GroupKFold
Best fold	Fold 4 (validation loss = 59.43)

After training, regimes are **reordered by total precipitation** so that Regime 1 is the wettest and Regime 4 is the driest circulation pattern.

Figures

Training Loss Curves

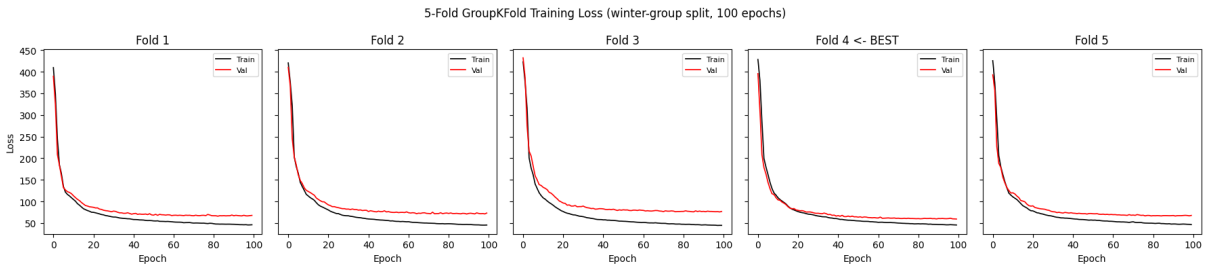


Figure 1: Training loss (black) and validation loss (red) per epoch for each of the 5 CV folds. Fold 4 achieved the lowest validation loss and was selected as the best model.

Z500 Cluster Centres

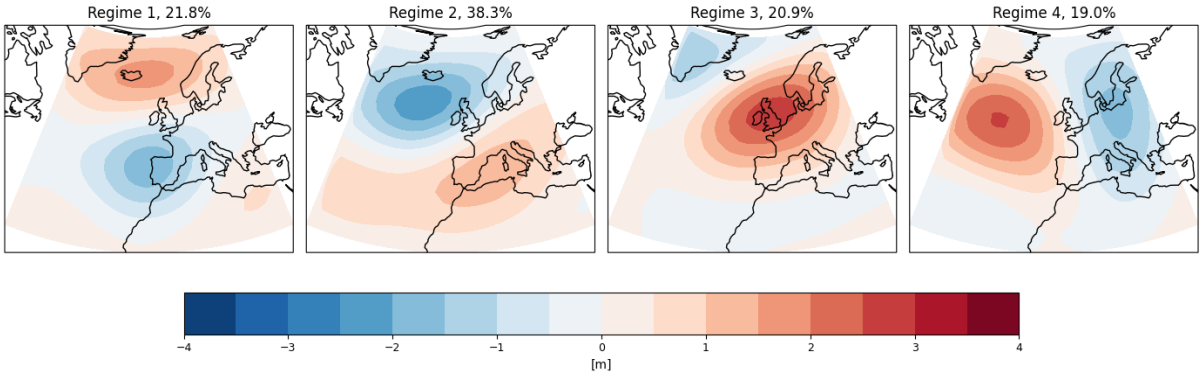


Figure 2: CMM-VAE: Z500 anomaly composite maps for each of the 4 regimes, ordered wettest to driest. Regime 1 corresponds to a deep Atlantic trough favouring moisture advection into Morocco. Regime 4 corresponds to blocking or ridging over the Atlantic.

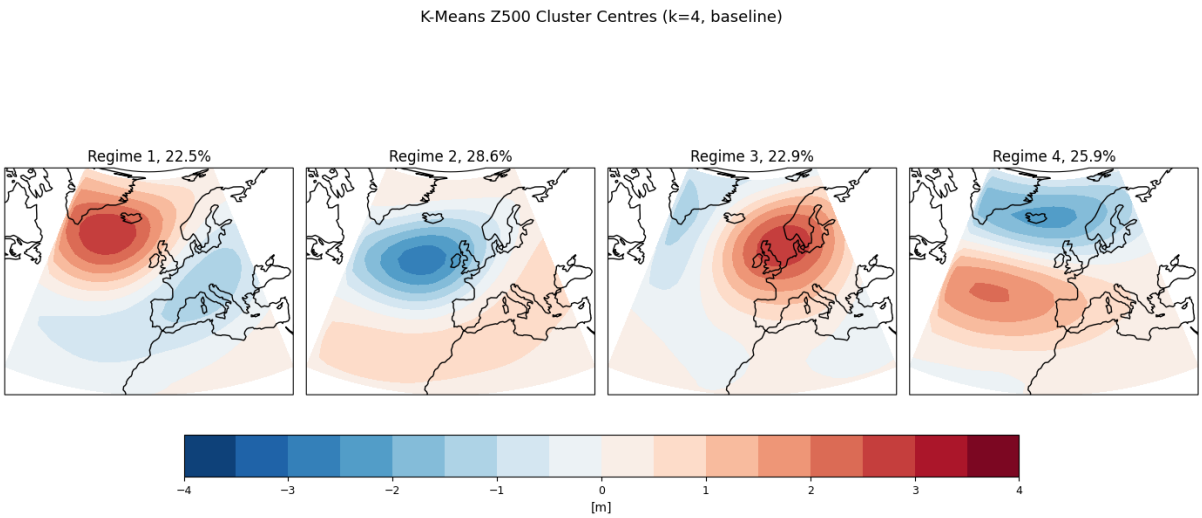


Figure 3: K-Means Baseline: Z500 cluster centre maps. The patterns are less tightly linked to Moroccan precipitation because the model clusters Z500 without any precipitation guidance.

Precipitation Odds Ratio Maps

The **odds ratio** at each grid point answers: *"How much more likely is extreme precipitation (above Q95) when this regime is active, compared to climatology?"*

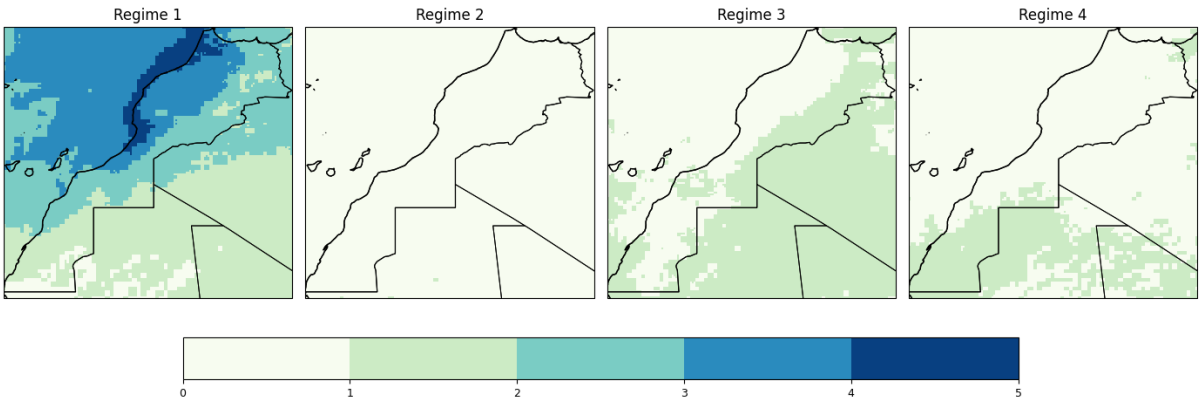
- Odds ratio > 1: regime increases likelihood of extreme rain
- Odds ratio < 1: regime suppresses extreme rain
- Odds ratio $= 1$: no signal

Three versions are computed for the CMM-VAE:

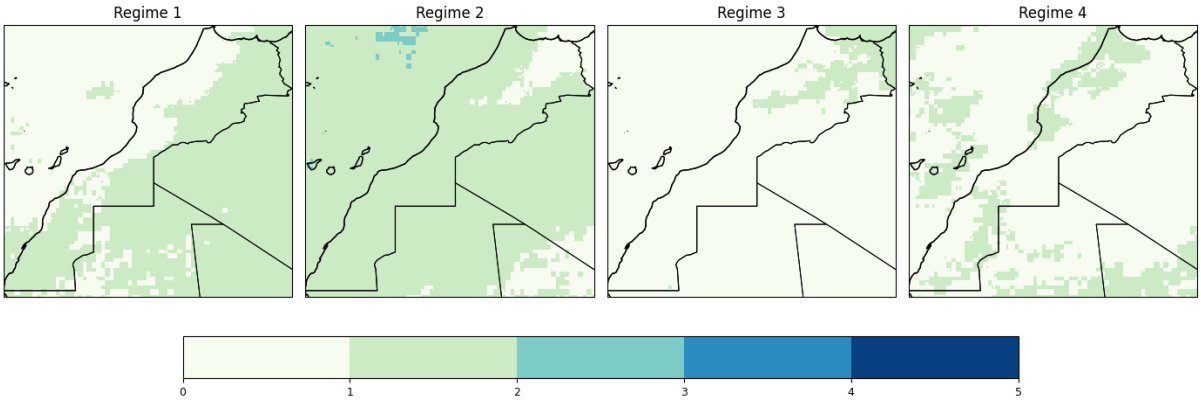
Table 4: Three precipitation odds ratio variants

Version	**Data used**	**Purpose**
Full period	All 42 winters	Overview / exploration
Training OOF	29 training winters (CV out-of-fold)	Check model learned real patterns
Holdout only	**13 unseen winters**	**True physical validation**

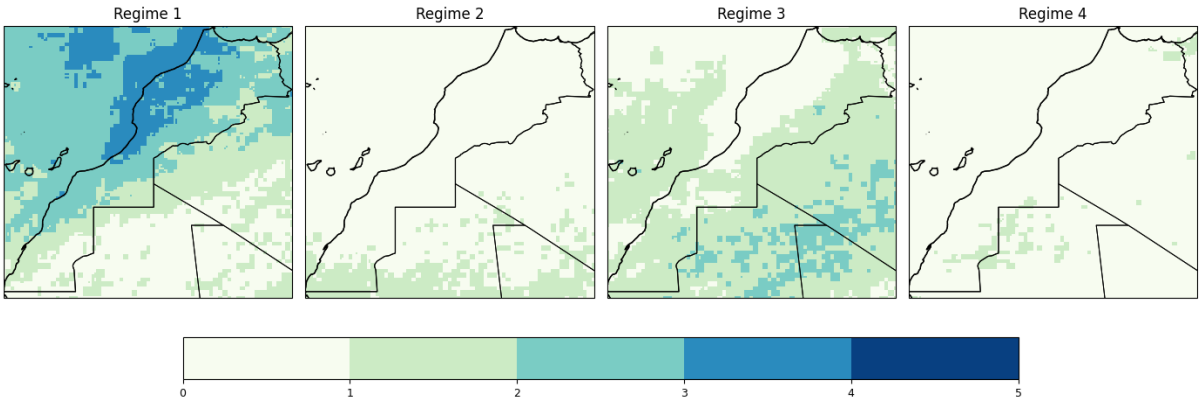
The **holdout odds ratio is the most important**: it shows whether the regime–precipitation link generalises to winters the model has never seen.



(a) Full period (1979–2020, all 42 winters). Optimistic view – model has seen most of these days during training.



(b) Training winters only – cross-validated out-of-fold predictions. Confirms the model learned physically consistent patterns without overfitting.



(c) **Holdout winters only (13 unseen winters) – primary physical validation.** Regime 1 shows elevated odds (> 1) over northern Morocco and the Atlas Mountains. Regime 4 is suppressed everywhere. Spatial coherence on unseen data confirms genuine generalisation.

Figure 4: **CMM-VAE** – Precipitation Q95 odds ratio maps for each of the 4 regimes (ordered wettest to driest), shown at three evaluation stages. The progression from (a) to (c) isolates the model's true out-of-sample skill: (a) is optimistic; (b) is honest for training data; (c) is the definitive test.

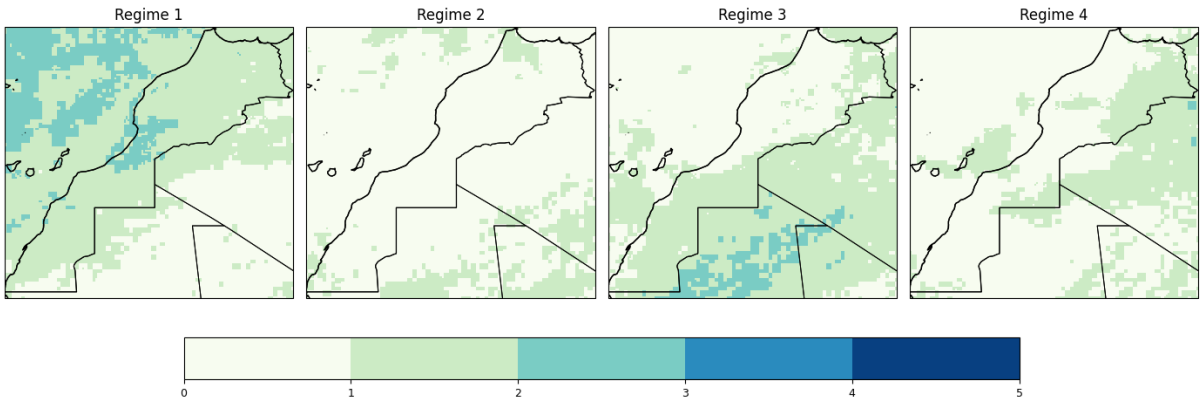


Figure 5: **K-Means Baseline** – Precipitation Q95 odds ratio on holdout winters only. Weaker and less spatially coherent signals compared to Figure 4(c), consistent with the lower skill scores. Unsupervised Z500 clustering without precipitation guidance fails to identify regimes with distinct precipitation signals.

Quantitative Results

Brier Skill Scores (BSS)

BSS > 0 means the model beats climatology. BSS $= 1$ is perfect. BSS < 0 is worse than climatology.

Table 5: Brier Skill Score comparison: K-Means baseline vs CMM-VAE

Metric	K-Means	CMM-VAE	Improvement
BSS 95th pct (CV)	0.0038	0.0293	+0.026
BSS cluster (CV)	0.0039	0.0112	+0.008
BSS 95th pct (holdout)	0.0279	**0.0818**	+0.054
BSS cluster (holdout)	0.0468	**0.2766**	+0.230
BSS tercile (holdout, mean)	0.0240	**0.0888**	+0.064

Key result: The BSS cluster score on the holdout set is the strongest signal. CMM-VAE achieves **0.277** versus the K-Means baseline's **0.047** – nearly **6× better** probabilistic skill at predicting which precipitation regime will occur on unseen winters.

Classification Accuracy on Holdout Set

Accuracy measures the fraction of holdout days where the model correctly predicts the precipitation cluster.

- **CMM-VAE:** uses (r_{holdout}) – the model directly outputs precipitation cluster probabilities.
- **K-Means baseline:** uses a majority-vote mapping – for each Z500 regime, the most frequent precipitation cluster in training becomes the prediction (derived from training data only, no leakage).

Table 6: Holdout set accuracy comparison

Metric	K-Means	CMM-VAE
Correct predictions	781 / 1172	**913 / 1172**
Accuracy	66.6 %	**77.9 %**
Climatology baseline	66.6 %	66.6 %
Gain over climatology	+0.0 pp	+11.3 pp

Table 7: Per-cluster accuracy breakdown (holdout set)

Precipitation Cluster	K-Means	CMM-VAE
Cluster 1 – dominant (781 days)	100.0 %	91.9 %
Cluster 2 (268 days)	0.0 %	**51.9 %**
Cluster 3 (68 days)	0.0 %	**38.2 %**
Cluster 4 (55 days)	0.0 %	**54.5 %**

Interpreting the K-Means Accuracy Result

The majority-vote mapping revealed a critical finding: all 4 Z500 regimes map to the **same** precipitation cluster (Cluster 1):

Z500 Regime	Predicted Precip Cluster
Regime 1 (517 training days)	Cluster 1
Regime 2 (776 training days)	Cluster 1
Regime 3 (603 training days)	Cluster 1
Regime 4 (723 training days)	Cluster 1

This means the K-Means baseline – which clusters Z500 *without* precipitation guidance – fails to produce Z500 groups that discriminate between different precipitation outcomes. Every regime predicts the dominant class, so accuracy equals climatology exactly (66.6 %). The baseline correctly classifies Cluster 1 days (the dry/near-normal majority) 100 % of the time but completely fails on Clusters 2, 3, and 4 (0 % each).

Key finding: Unsupervised Z500 clustering alone is insufficient to predict Moroccan precipitation regimes. The CMM-VAE's *joint* learning of circulation and precipitation is what enables it to identify regimes genuinely linked to different precipitation outcomes – including rare wet events (Clusters 2–4) that the baseline misses entirely.

Summary and Conclusions

Table 8: Summary of key findings

Aspect	Finding
Data leakage	Eliminated – all 13 holdout winters are completely unseen during training; KMeans labels and Q95 threshold derived exclusively from training data.
Accuracy	CMM-VAE: **77.9 %** (+11.3 pp over climatology). K-Means: 66.6 % (= climatology, +0.0 pp).
BSS cluster (holdout)	CMM-VAE: **0.277** vs K-Means: 0.047 – nearly **6**× better probabilistic skill.
BSS 95th pct (holdout)	CMM-VAE: **0.082** vs K-Means: 0.028 – better at forecasting extreme rain days.
K-Means failure mode	All 4 Z500 regimes map to the same precipitation cluster; the baseline learns nothing about precipitation discrimination.
CMM-VAE discriminative power	Correctly predicts minority clusters (Cluster 2: 51.9 %, Cluster 3: 38.2 %, Cluster 4: 54.5 %) that the baseline misses entirely.
Physical realism	Holdout odds ratio maps show spatially coherent wet/dry signals over northern Morocco and the Atlas Mountains, consistent with known orographic and large-scale dynamics.
Cross-validation	5-fold GroupKFold respects winter boundaries; best model = Fold 4 (validation loss = 59.43).
Robustness	Positive skill on completely unseen winters confirms the model generalises beyond its training period.

Notebooks

Notebook	Purpose
`cmmvae_mswep_djf_winter_groups.ipynb`	Main CMM-VAE model with leakage-free winter-group
`baseline_kmeans_djf.ipynb`	K-Means Z500 baseline with identical split for fair comp